Convolutional Neural Networks (CNNs)

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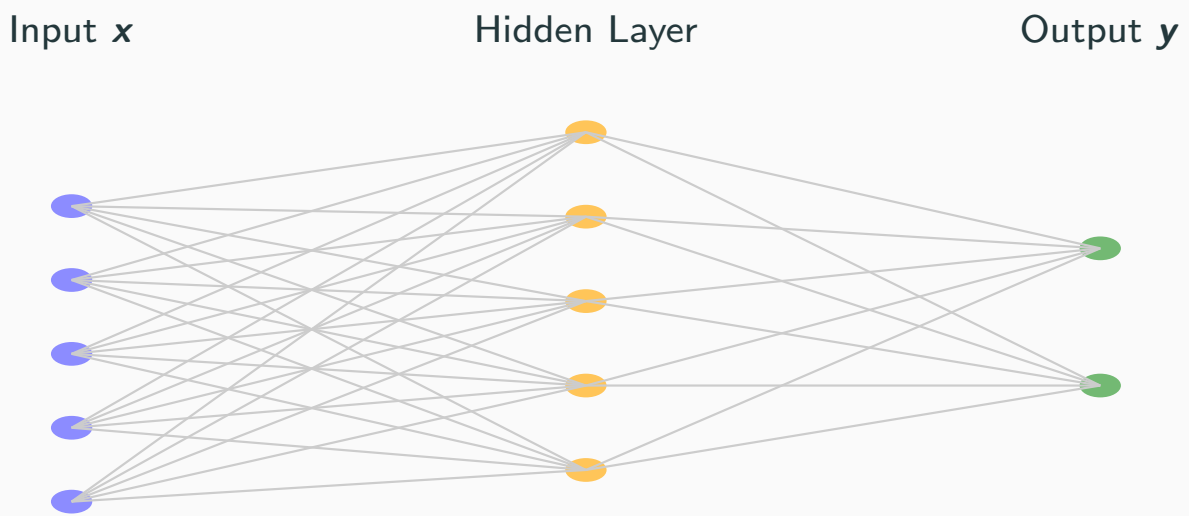
Quick Recap: a Basic Neural Network (MLP)

- A (fully-connected) layer maps an input vector $\boldsymbol{x} \in \mathbb{R}^D$ to $\boldsymbol{y} \in \mathbb{R}^K$:

$$\boldsymbol{y} = \sigma(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b})$$



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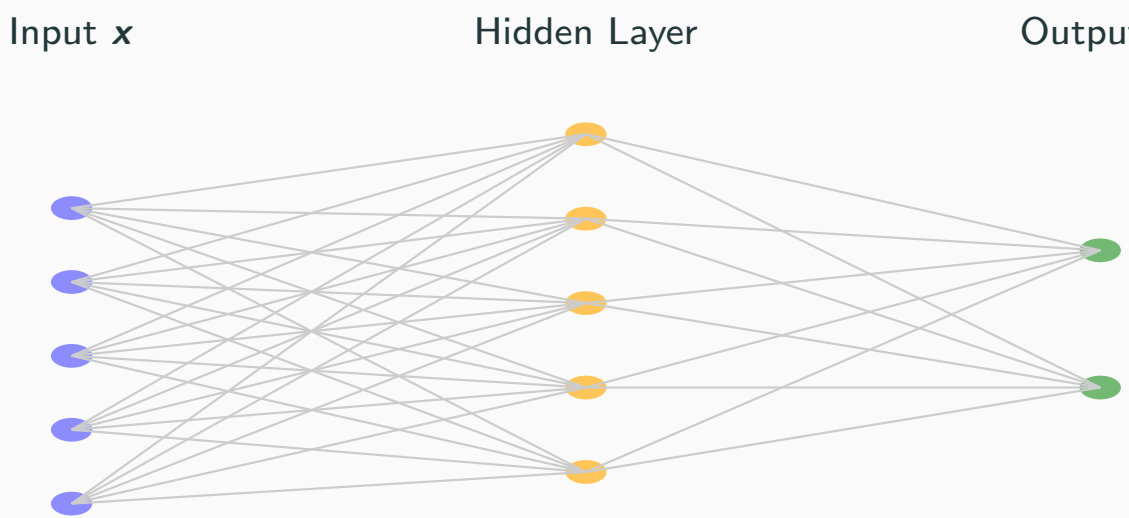

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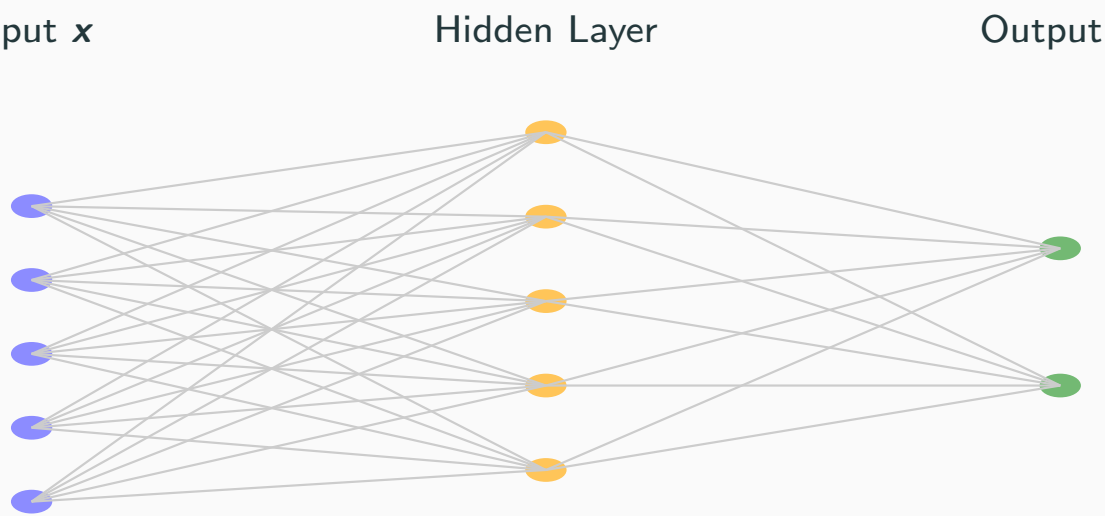
- **Parameters:** $(d \times M + M \times K)$ weights + $(M + K)$ biases. Where M is the number of hidden units (nodes).
- MLPs treat inputs as 1D vectors (no explicit spatial structure).



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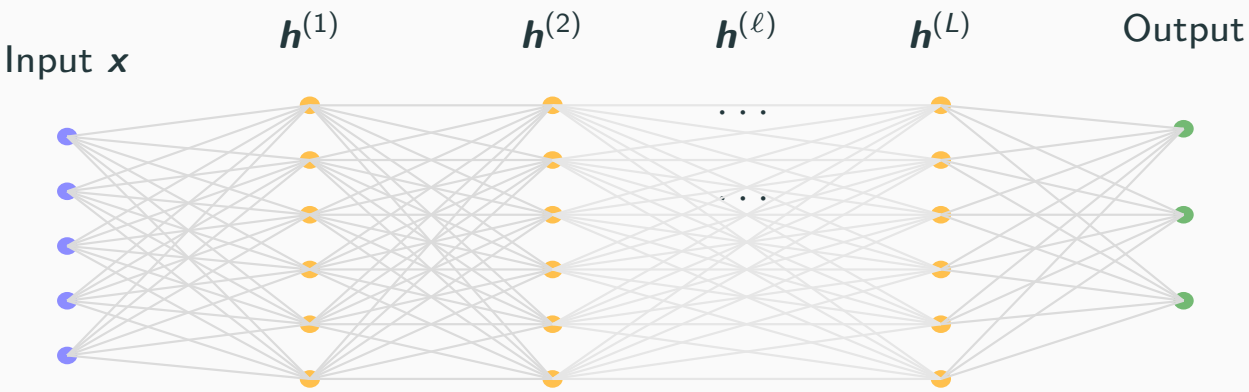

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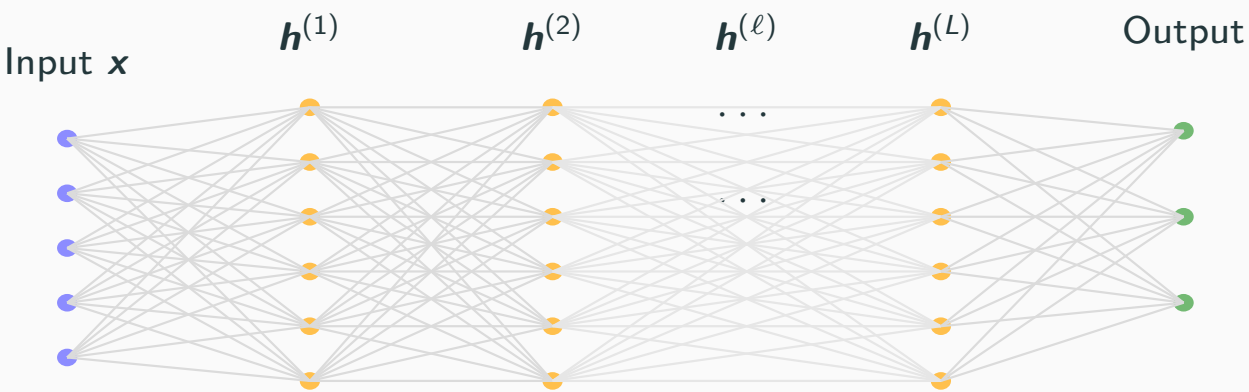

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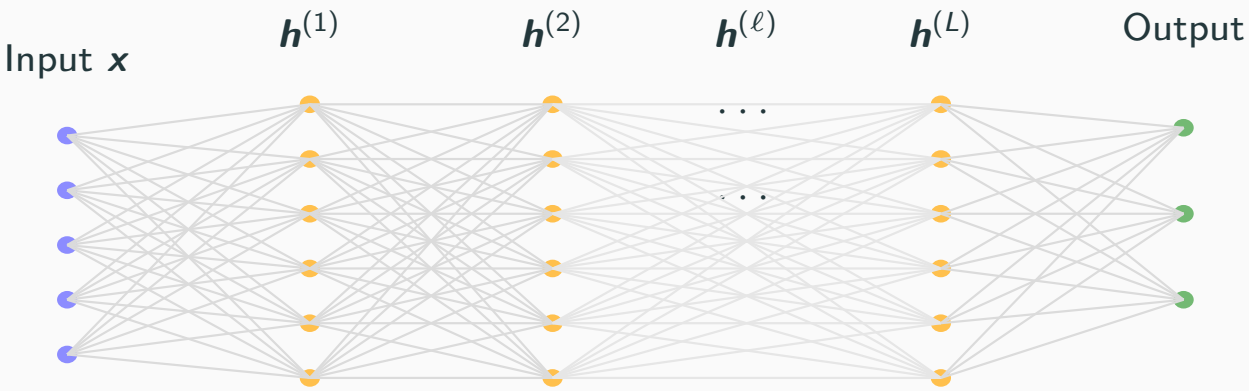
- Output layer (example for classification):
  $\boldsymbol{z}^{(L+1)} = \sigma\left(\boldsymbol{W}^{(L+1)} \boldsymbol{h}^{(L)} + \boldsymbol{b}^{(L+1)}\right)$; here σ must be the softmax function.



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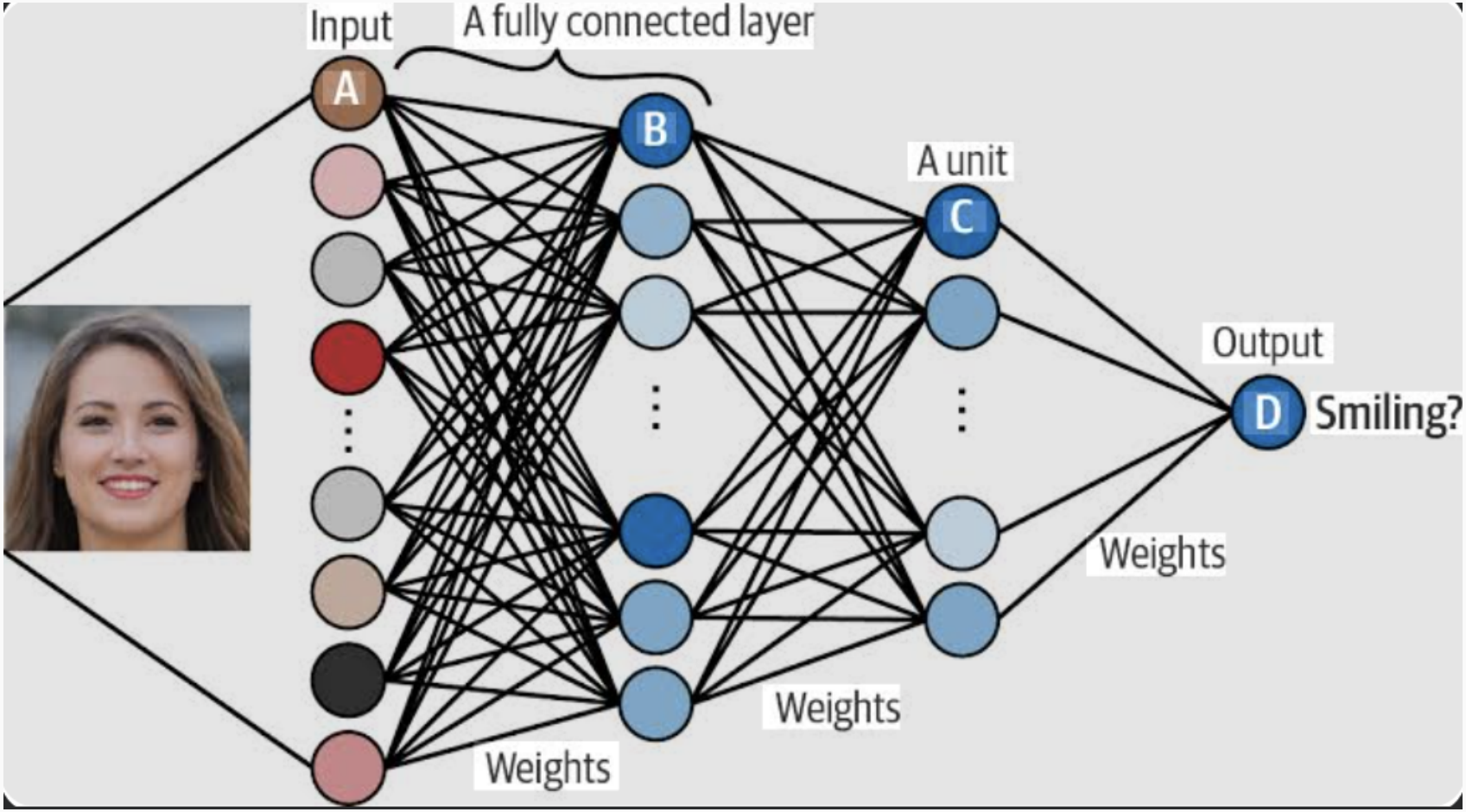
- These pixel values are just numbers used as input features in ML/CNNs.

MLPs for Image Data: Just Flatten an Image and use an MLP



Why not Just Flatten an Image and use an MLP? (Parameter Explosion) II

Example: RGB image $64 \times 64 \times 3$ flattened to $d = 12{,}288$.

- Dense layer with $m = 1{,}024$ hidden units has:

$$\#\text{params} = d\,m + m = 12{,}288 \cdot 1{,}024 + 1{,}024 \approx 12.6 \text{ million}$$

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- **Memory/latency:** large dense matrices are expensive (GPU/TPU memory + bandwidth).

What Structure do Images Have That MLPs Ignore?

- **Locality**: nearby pixels are more related than far-away pixels.
- **Translation**: the same pattern (edge/texture) can appear anywhere in the image.
- **Stationarity**: low-level statistics (edges/textures) are similar across locations.
- **Compositionality**: edges $\rightarrow$ corners $\rightarrow$ parts $\rightarrow$ objects.
- **Multi-scale structure**: relevant patterns exist at multiple spatial scales.

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CNNs encode these as **inductive biases** via:

- **Local connectivity** (small receptive fields),
- **Weight sharing** (same kernel everywhere),
- **Downsampling** (pooling/striding) for multi-scale features,
- **Hierarchy**: deeper layers aggregate simpler features into more complex ones.

Note: Convolutions are **translation equivariant** (shifting the input shifts the feature map). Pooling and global averaging help build **approximate invariance**.

Convolutional Neural Networks (CNNs)

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From MLPs to CNNs: the key Idea

- A **CNN** is a neural network designed for grid-structured data (images, spectrograms, time-series).
- Replace dense connections with **convolutions**: a small kernel slides across the input.
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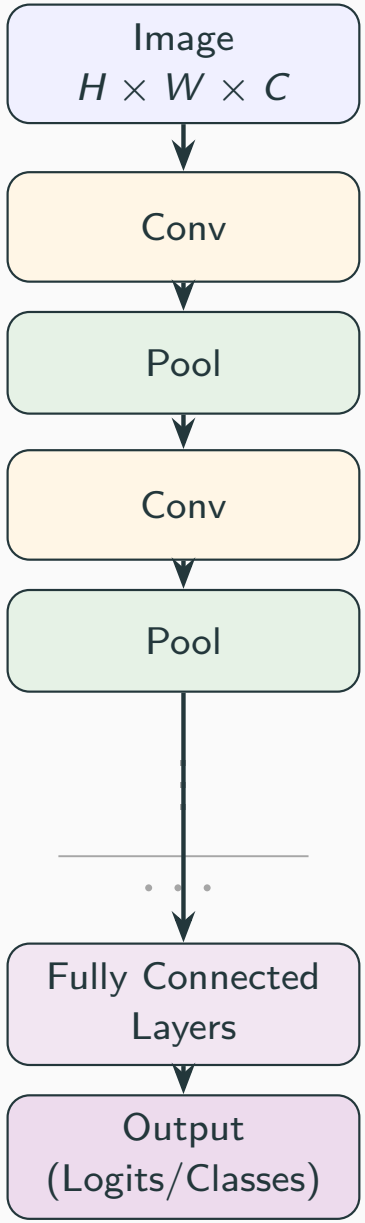
Parameter comparison (typical):

- Dense layer (flattened input): $\#\text{params} \approx (H\,W\,C) \cdot m$.
- Conv layer: $\#\text{params} = K\,(k\,k\,C + 1)$ (independent of H, W).

- Benefits: **fewer parameters**, **translation equivariance**, and **hierarchical feature learning**.
- Typical pipeline: **Conv** $\rightarrow$ **Nonlinearity** $\rightarrow$ (optional) **Normalization** $\rightarrow$ (optional) **Pooling/Stride** $\rightarrow$ repeat.
- End: **global average pooling** or **fully connected** head for predictions.

A Standard CNN Block (and why it Works)

- **Convolution layer**: learn local feature detectors (edges, textures, parts).



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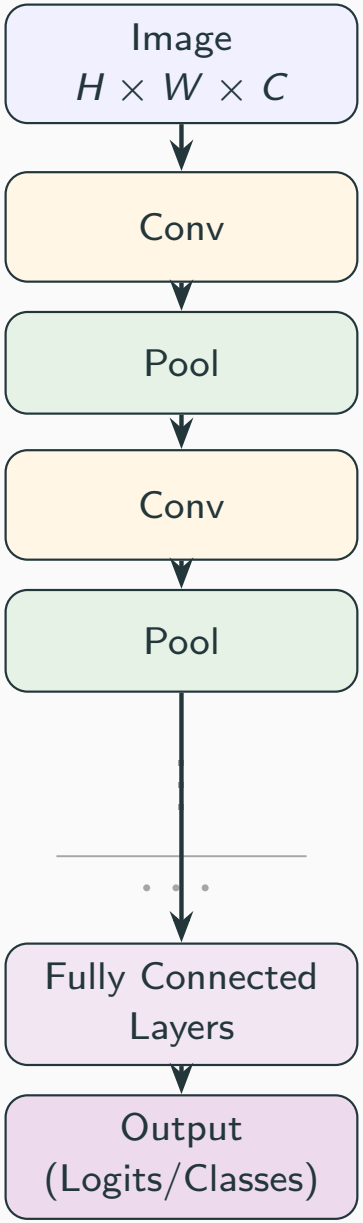

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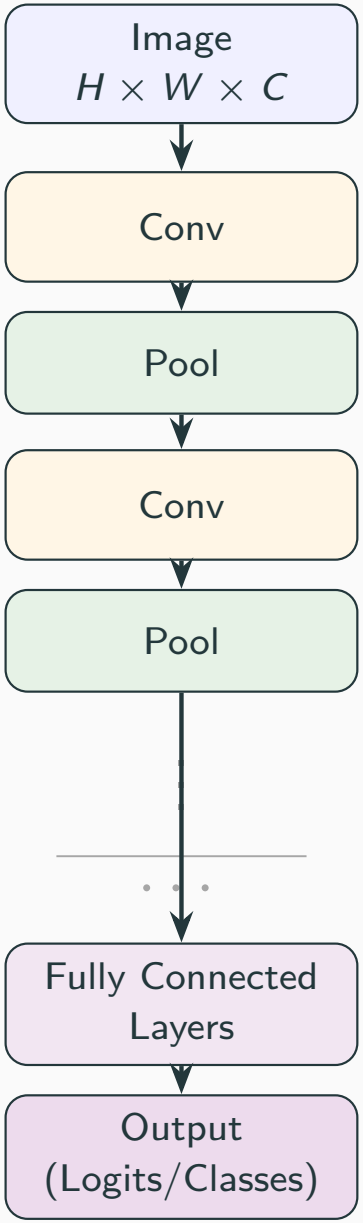

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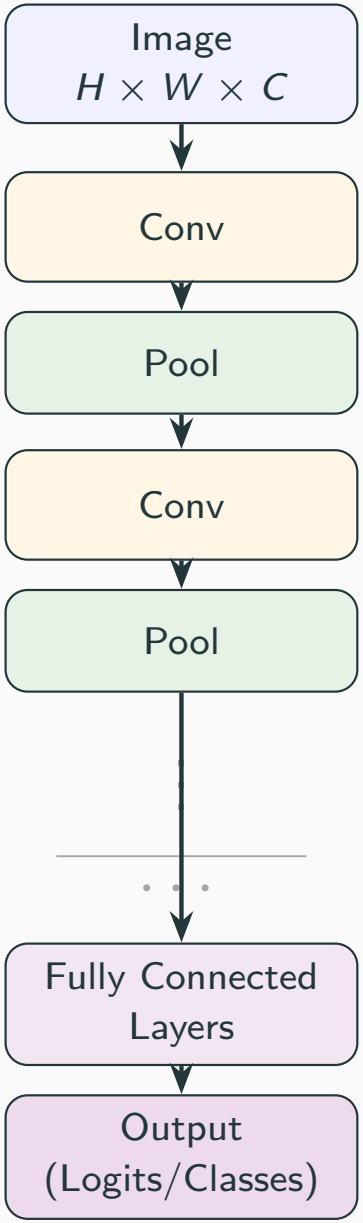

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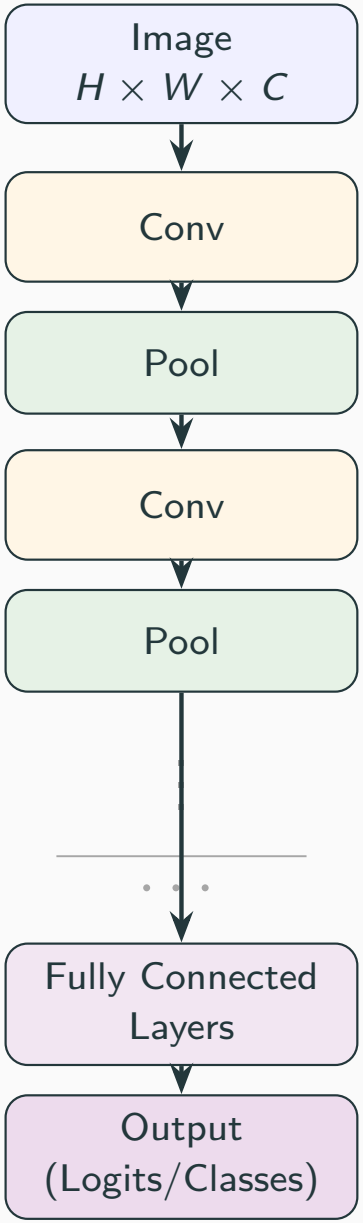

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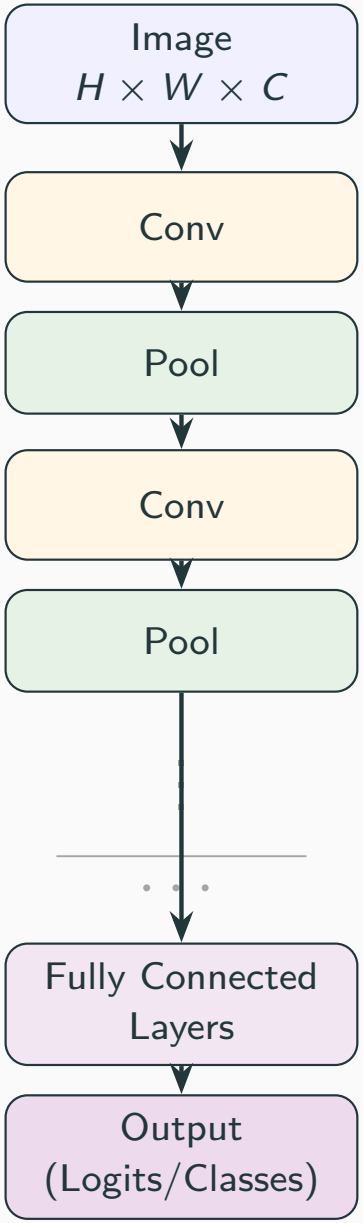
- **Fully Connected Layers**: fully-connected (FC) or global pooling $\rightarrow$ classifier.



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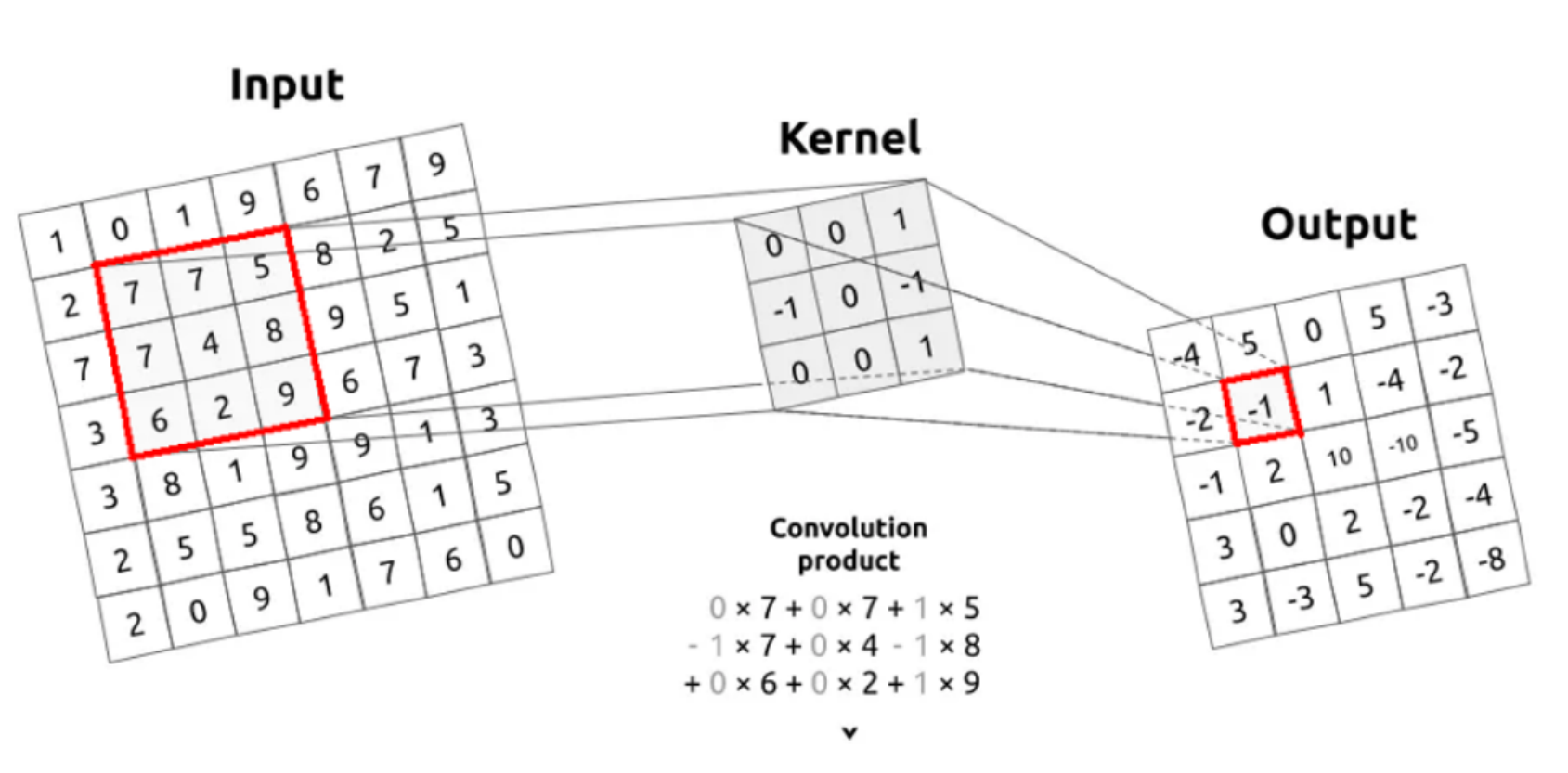
What is Convolution? (Intuition)

- A small filter scans across the image
- Detects patterns like edges, textures
- Same filter applied everywhere



Convolution Operation

$$Y[i,j] = \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} W[u,v]\, X[i+u, j+v]$$

- W: filter (kernel), learned during training
- X: input image (or previous feature map)
- Y: output feature map (one map per filter)
- Intuition: each output pixel is a *weighted sum* of a local patch

Learn more from: https://developer.nvidia.com/discover/convolution

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- **Interpretation:** A channel captures one type of information or feature across the spatial/temporal dimensions.
- **Special cases:**
  - **1D data (signals, time series):** $X \in \mathbb{R}^{L \times C}$
    Channel = 1D signal
  - **2D data (images):** $X \in \mathbb{R}^{H \times W \times C}$
    Channel = 2D map (e.g., RGB planes)
  - **3D data (videos, volumetric):** $X \in \mathbb{R}^{D \times H \times W \times C}$
    Channel = 3D volume

Filters (Kernels): How Channels are Produced i

- A **filter/kernel** detects local patterns by sliding over the input
- **Key idea:**
  - Each filter spans **all input channels**
  - **One filter $\Rightarrow$ one output channel**
  - K **filters** $\Rightarrow$ K **output channels**
- **1D Convolution (Time Series)**
  - Input: $X \in \mathbb{R}^{L \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times C}$
  - Output (one channel):

$$Y[:, m] = \sum_{c=1}^{C} W^{(m)}[:, c] * X[:, c]$$

  - Using K filters $\Rightarrow$ $Y \in \mathbb{R}^{L' \times K}$

Filters (Kernels): How Channels are Produced ii

- **2D Convolution (Images)**
  - Input: $X \in \mathbb{R}^{H \times W \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times k \times C}$
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$$Y[:, :, m] = \sum_{c=1}^{C} W^{(m)}[:, :, c] * X[:, :, c]$$

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- **3D Convolution (Videos / Volumes)**
  - Input: $X \in \mathbb{R}^{D \times H \times W \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times k \times k \times C}$
  - Output (one channel):

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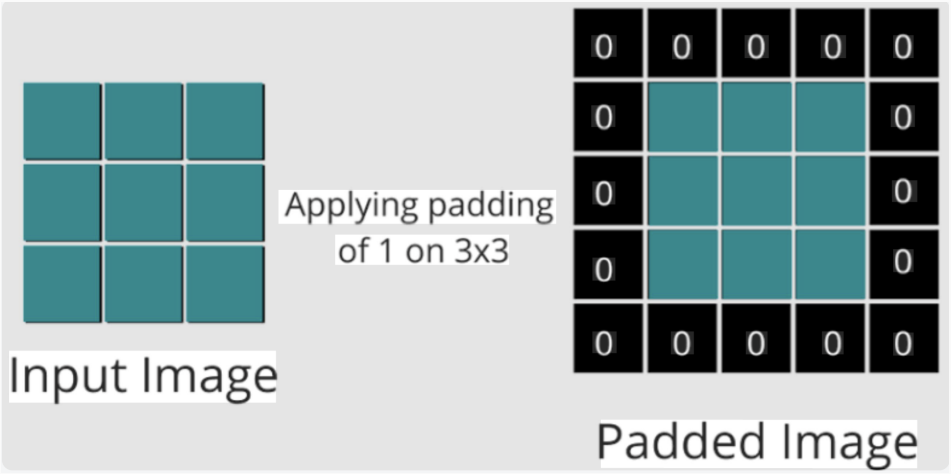
Padding in CNNs: When and Why?

- **Padding** adds border pixels (usually zeros) so convolution can be applied near edges



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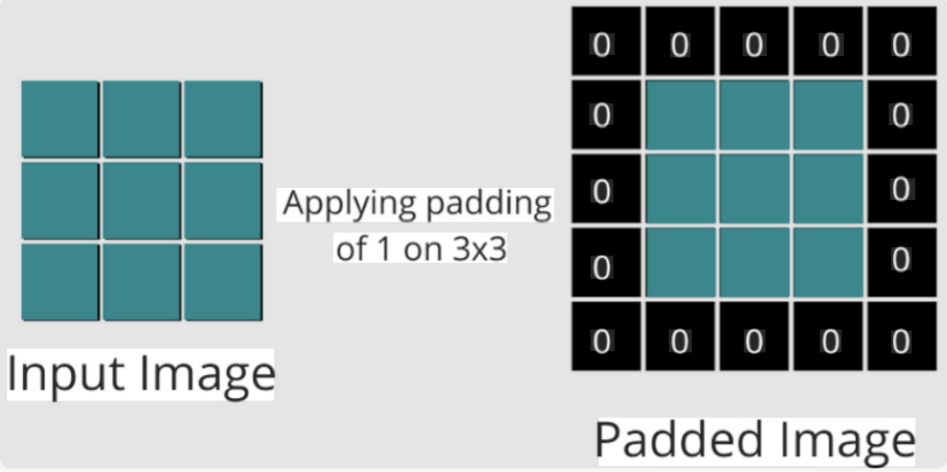

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  - preserves information at image borders (otherwise borders are seen fewer times)
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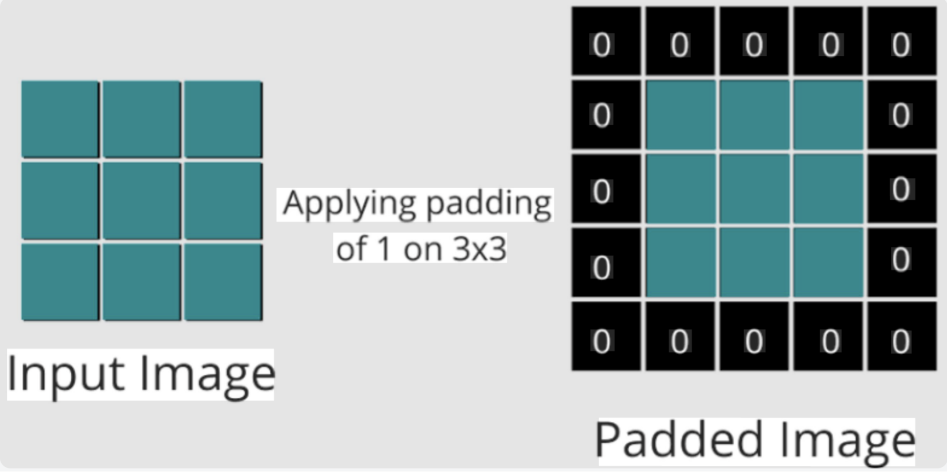

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- Two types of padding:
  - **Valid** padding: $P = 0$ (no padding) $\Rightarrow$ output shrinks
  - **Same** padding: choose P to roughly preserve size

Activation Functions in CNNs: Why and When?

- **Why do we need them?**
  - Without activation functions, the network is just a **linear model**
  - Stacking linear layers $\Rightarrow$ still linear (no extra power)
  - Activations introduce **non-linearity** $\Rightarrow$ can learn complex patterns

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- **Leaky ReLU**

$$\text{LeakyReLU}(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}, \qquad \alpha > 0 \qquad \text{is a small constant}$$

  - Fixes "dead neuron" problem in ReLU
  - Allows small gradient when $x < 0$
- **When are they applied?**
  - Applied **after every convolution layer**: Typically `Conv` $\rightarrow$ `ReLU`

Batch Normalization: Why and How?

- **Why do we need BatchNorm?**
  - Deep networks are hard to train due to:
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$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta$$

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- **Where is it applied?**
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Key Hyperparameters in Convolution

- **Kernel size** ($k_h \times k_w$): size of filter window (e.g., 3×3)
- **Stride** (S): step size when sliding the filter (e.g., $S = 2$ downsamples)
- **Padding** (P): zeros added at borders to control output size
  - **Valid**: no padding ($P = 0$)
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- **Number of filters** (K):
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Output Size Formula

$$H' = \left\lfloor \frac{H + 2P - k_h}{S} \right\rfloor + 1, \qquad W' = \left\lfloor \frac{W + 2P - k_w}{S} \right\rfloor + 1$$

$$\text{Output: } X' \in \mathbb{R}^{H' \times W' \times K}$$

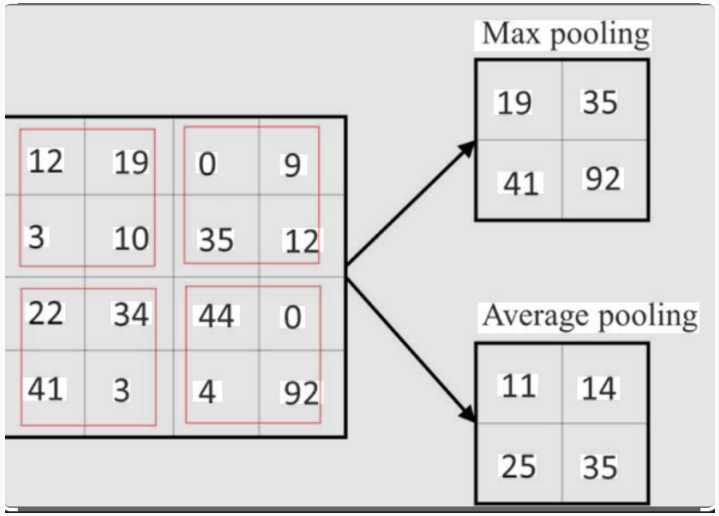
- Example: $H = W = 32,\ k = 3,\ S = 1,\ P = 1 \Rightarrow H' = W' = 32$ (same)

Why Pooling?

- **Reduces spatial size**:
  - From $H \times W \to H' \times W'$
  - Reduces computation and memory
- **Provides translation invariance**:
  - Small shifts in input do not change output much
- **Increases receptive field**:
  - Each pooled value summarizes a larger region
- **Reduces overfitting**:
  - Acts as a form of downsampling / regularization

Pooling Types (and When to Use Them)

- **Max Pooling**
  - Takes the maximum value in each window
  - **Important when:** you care about *whether a feature exists* (edge/texture presence), and want robustness to small shifts
  - Common in early CNNs and in feature-extraction backbones
- **Average Pooling**
  - Takes the mean value in each window
  - **Important when:** you want *smoother* / less noisy activations (e.g., low-texture regions, denoising-like effects)



How Pooling Works

- Applied **independently on each channel**
- For input:

$$X \in \mathbb{R}^{H \times W \times C}$$

- A window of size $k \times k$ slides with stride s
- Output:

$$Y \in \mathbb{R}^{H' \times W' \times C}$$

- **Important:**
  - Number of channels C remains unchanged
  - Only spatial dimensions are reduced

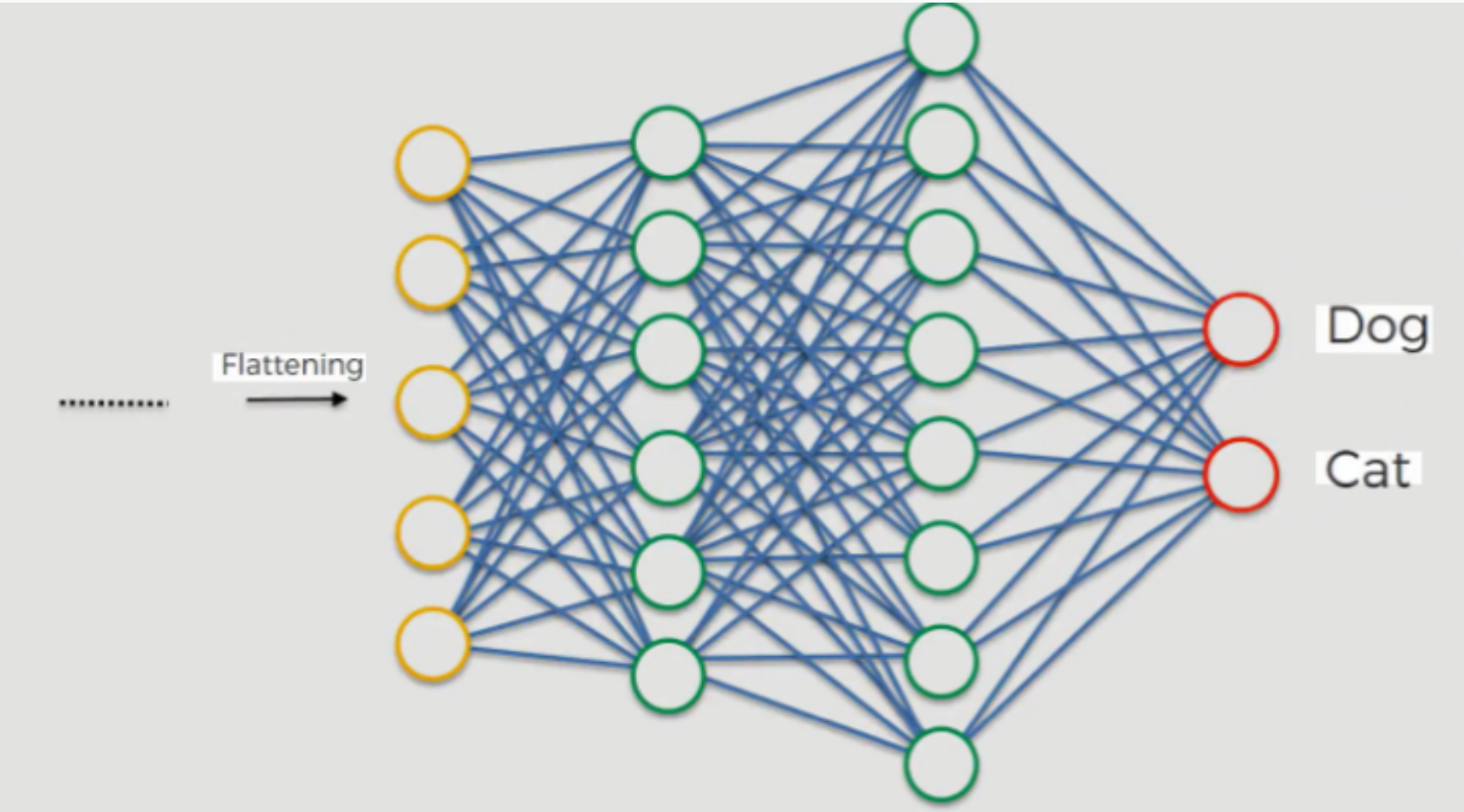
Fully Connected (Dense) Layers i

- A **Fully Connected (FC)** layer connects every input unit to every output unit
- **Role in CNNs:**
  - Placed near the **end of the network**
  - Combines all extracted features **globally**
  - Performs **final prediction / classification**
- **Input:**
  - Feature maps are **flattened** into a vector

$$X \in \mathbb{R}^{H \times W \times C} \rightarrow \mathbf{x} \in \mathbb{R}^{n}$$

    where $n = HWC$
- **Interpretation:**
  - Learns **global combinations** of features
  - Acts like a standard **neural network layer**
  - Has **large number of parameters** (can cause overfitting)

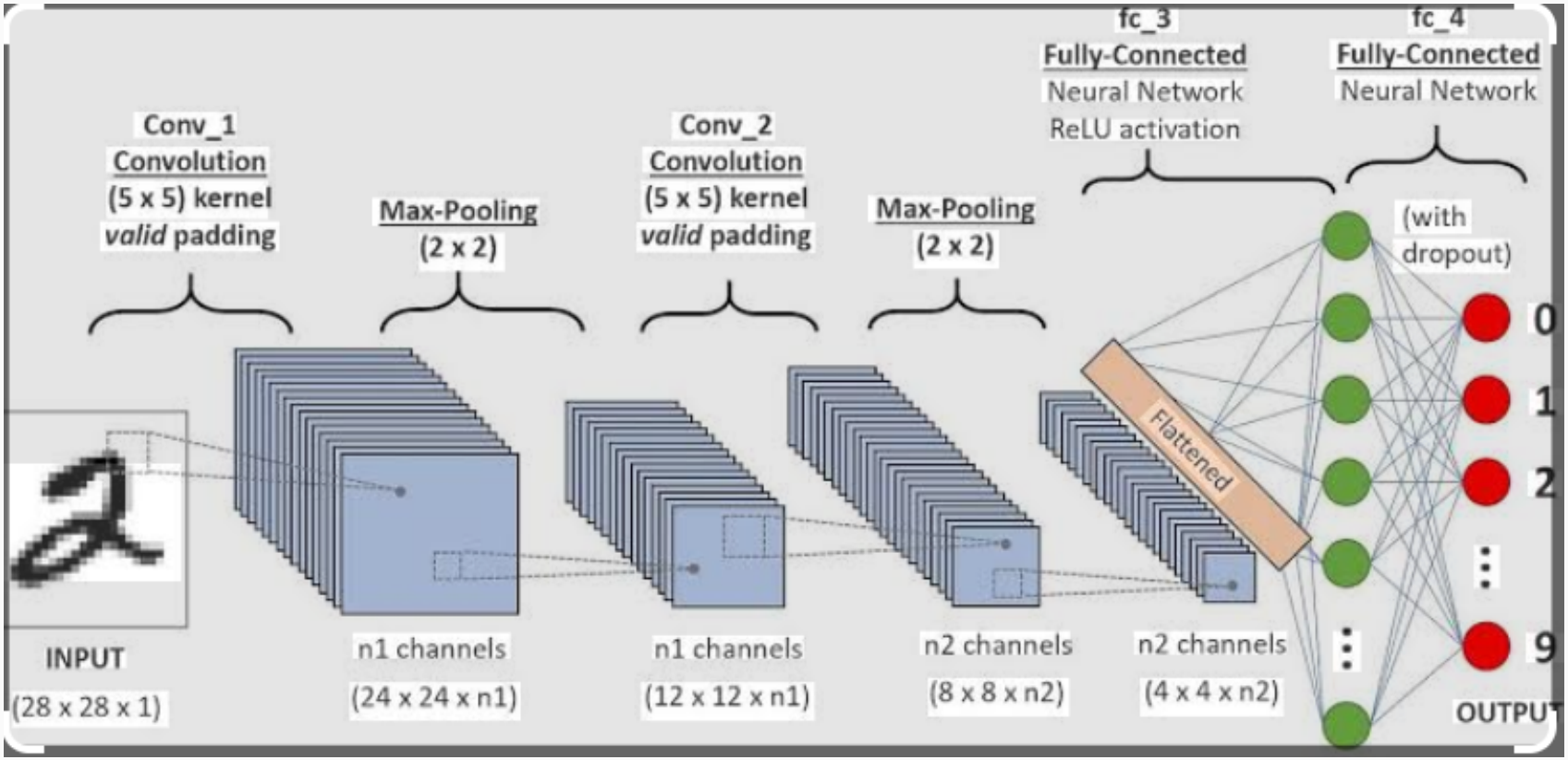
Fully Connected Layers ii



Agenda

One Example of CNN Architecture

Input $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ Pool
$\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ Pool
$\rightarrow$ Fully Connected $\rightarrow$ Softmax (for classification)



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- **Step 4: Parameter Update**
  - Gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

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CNNs Lab

Two examples: Look at the following GitHub link: https://github.com/yadavrishikesh/Masterclass-V2-2026/blob/main/module2/code and then the following notebook

1. **CNNs_grayScaleImage.ipynb**: CNNs for digit recognition (example of gray scale image)
2. **CNNs_for_RGBImage_Data.ipynb.ipynb**: CNNs for object detection (Example of of RGB channel)